

Conservation implications of inaccurate estimation of cryptic population size

T. E. Katzner¹, J. A.R. Ivy^{2,*}, E. A. Bragin³, E.J. Milner-Gulland⁴ & J. A. DeWoody²

¹ Department of Conservation and Field Research, National Aviary, Pittsburgh, PA, USA

² Department of Forestry and Natural Resources, Purdue University, West Lafayette, IN, USA

³ Science Department, Naurzum National Nature Reserve, Kostanay Oblast, Naurzumski Raijon, Karamendy, Kazakhstan

⁴ Division of Biology and Centre for Population Biology, Imperial College London, Silwood Park Campus, Ascot, Berkshire, UK

Keywords

non-invasive monitoring; cryptic population size; demographic models; non-breeding population density; population monitoring; imperial eagle; *Aquila heliaca*.

Correspondence

Todd Katzner. Current address: Division of Forestry and Natural Resources, West Virginia University, PO Box 6125, Morgantown, WV 26506, USA.
Email: todd.katzner@mail.wvu.edu

Editor: Steve Redpath

*Current address: Department of Animal Collections, Zoological Society of San Diego, PO Box 120551, San Diego, CA 92112 0551, USA.

Received 4 October 2010; accepted 1 January 2011

doi:10.1111/j.1469-1795.2011.00444.x

Abstract

Estimating population size is central to species-oriented conservation and management. However, in spite of recent development in monitoring protocols, there are gaps in our ability to accurately and quickly estimate numbers of individuals present, especially for the cryptic and often non-breeding components of structured vertebrate populations. Yet knowing the size and growth trajectory of all stage classes of a population is critical for species conservation. Here we use data from 2 years of non-invasive genetic sample collection from the cryptic, non-breeding component of an endangered bird of prey population to evaluate the impact of variability in population estimates on demographic models that underpin conservation efforts. A single non-invasive sample collection in 2003 conclusively identified 47 individual non-breeding imperial eagles, 2.8 times more than were visually counted. In 2004, our comprehensive genetic and observational analyses determined that 414 imperial eagles ($n = 308$ non-breeders + 68 territory holders + 38 chicks) were present. This estimate was 326% larger than the 127 birds visually observed ($n = 21$ non-breeders + 68 territory holders + 38 chicks) and 265% larger than the population size predicted by demographic models with the same number of breeders ($n = 156 \pm 7.2; \pm \text{SE}$). Our study builds on a body of work that demonstrates that conventional visual estimation of cryptic components of structured populations may not always be effective. Furthermore, we show that reliance on those estimates can result in inaccuracies in the demographic models that are often the foundation for subsequent conservation action.

Introduction

Estimating population size is central to species-oriented conservation and management (Yoccoz, Nichols & Boulinier, 2001; Martin, Kitchens & Hines, 2007). Although monitoring strategies have been the subject of extensive recent research, the cryptic and often non-breeding components of structured populations are almost never included in population estimates. However, effective conservation requires solid estimates for all stage classes of a population. This is especially true in highly structured populations of vertebrates, where transient non-breeders may form an essential buffer against fluctuations in population size and potentially compete with breeders for scarce resources (Doak, 1995). Few studies have attempted to estimate numbers of non-breeding birds and many of those that do involve substantial effort. Two such studies required 10–20 year commitments and individual tracking of hundreds of birds (Kenward, Marcström & Karlbom, 1999; Kenward, *et al.*, 2000; Newton & Rothery, 2001).

In 2 separate years we counted, visually and non-invasively with DNA fingerprints, the number of eastern imperial eagles *Aquila heliaca* and white-tailed sea eagles *Haliaeetus albicilla* at the Naurzum National Nature Reserve in north-central Kazakhstan. The Naurzum Reserve supports the world's highest known breeding density of globally vulnerable imperial eagles as well as many non-territorial, non-breeding eagles that roost communally among defended territories. These birds represent an unknown proportion of the total regional eagle population, yet this cryptic component of the population has proven almost impossible to count visually. Here we genetically evaluate non-invasively collected feather samples from this component of the population to evaluate the impact of variability in traditional population estimates on demographic models that underpin current conservation efforts in the region. Specifically, (1) we expand upon previous study (Rudnick *et al.*, 2008) to reinforce the value of non-invasive genetic monitoring; (2) we evaluate how the use of input parameters from traditional versus non-invasive

genetic data sources impacts interpretation of demographic models.

Materials and methods

Study area and field studies

The Naurzum National Nature Reserve is located in the Kostanay Oblast (Kostanay administrative region) of north-central Kazakhstan (51°N, 64°E; Fig. 1). In addition to white-tailed sea and imperial eagles, the reserve also supports substantial breeding populations of steppe *Aquila nipalensis* and golden eagles *Aquila chrysaetos*. The reserve is dominated by a steppe matrix of sandy and mixed soils with feather (*Stipa* spp.) and bunch grasses or denser clay soils with low sagebrush (*Artemisia* spp.) and other nutrient-rich shrubs and grasses. Interspersed within the steppe matrix are forested areas of birch (*Betula* spp.), pine *Pinus sylvestris* and aspen (*Populus* spp.). Imperial eagles are migratory, arriving at the reserve in March, incubating from April into May and fledging chicks in late July and early August. Sea eagles may be resident year round and their breeding cycle starts about 2–3 weeks earlier than imperial eagles. Eagles nest primarily in the forests and the steppe matrix supports diverse prey assemblages.

Non-territorial eagles, nearly all in pre-adult plumage (i.e. juveniles and subadults), are regularly observed roosting communally between breeding territories. In 2003 and 2004, the most heavily used communal roosting areas were spread among four clusters of five to 15 trees each, distributed in the centre of the Naurzum region of the reserve (Fig. 1). The mean distance between neighbouring roosts was 1.02 km (range 0.43–1.82 km).

We evaluated records from co-author E. A. B.'s field notebooks since 1978 and recorded the maximum number of individual birds seen leaving the roost by a single observer during opportunistic observations at these roosts. These observations occurred between 20 May and 13 July, not always on the same days each year. Adult birds were also monitored visually on territories from 1978 to 2004 as reported in Katzner, Milner-Gulland & Bragin (2007), and genetically from 1998 to 2003 as reported in Rudnick *et al.* (2005).

Breeding and non-breeding imperial eagles shed many feathers at this time of the year and we collected newly moulted and naturally shed eagle feathers from beneath two of the four roost tree clusters once in 2003 (144 feathers) and four times at all four roost clusters over 20 days in July 2004 (1822 feathers; Rudnick *et al.*, 2008). Feathers moulted in previous years show heavy weathering on the shaft and plume and are readily discernable from those more recently moulted; heavily weathered feathers were not collected in this study. Feathers were collected by hand, cut to a 5–10 cm piece that included the tip, and stored at room temperature in paper envelopes.

Laboratory genetic analyses

DNA was isolated from samples as described elsewhere (Rudnick *et al.*, 2005). A negative control containing no eagle tissue was included in each set of DNA extractions, and a no-template negative control was included in each PCR set. Previous work on eastern imperial eagles employed nine microsatellite markers with a global, across loci, genotyping error rate of <1.5% under the same sample collection methods and analysis techniques used here (Rudnick *et al.*, 2005). A subset of seven microsatellites that provided the most robust PCR amplifications (IEAAG-15, Aa02, Aa35, Aa36, Aa39, Aa43 and Aa49) was used for the current analyses (Rudnick *et al.*, 2005). All samples were genotyped as in Rudnick *et al.* (2005), and we used these markers for individual identification of both imperial and white-tailed sea eagles.

Microsatellite profiles from feathers were used to group genetically identical samples. To limit errors associated with non-invasively collected samples (e.g. allelic dropout, null alleles), we used a method of 'sample culling' similar to that suggested by Paetkau (2003) that ensures high quality data. Any sample that failed to amplify at a minimum of two of the first four loci genotyped was immediately discarded. Samples that were not discarded were genotyped at three additional loci. Any imperial eagle sample that failed to amplify at a minimum of five loci was discarded. For a detailed description of error control technique, see Rudnick *et al.* (2008).

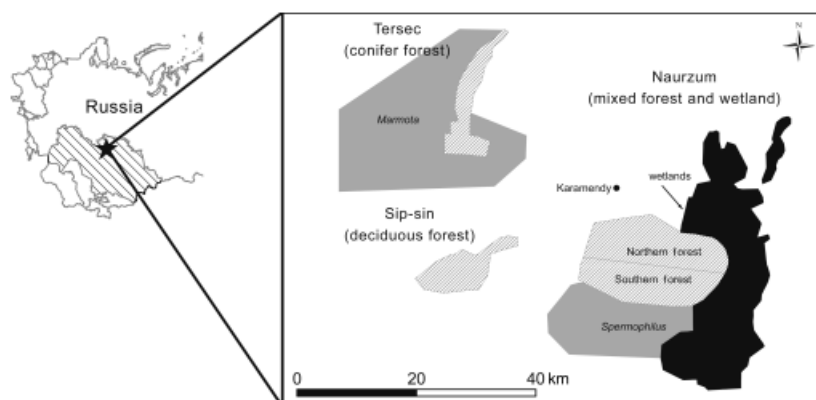


Figure 1 Map of the Naurzum National Nature Reserve, Kazakhstan, showing its location in the former Soviet Union, the three administrative and four biotic forest regions of the reserve, and the approximate location of marmot and ground squirrel colonies that are the main prey of local eagles. The imperial eagle *Aquila heliaca* roost sampled for this research is located in the approximate centre of the 'Southern Forest' of the Naurzum region.

In 2003, we identified eagle species based on microsatellite profiles, as three loci (Aa02, Aa35 and Aa36) harboured different alleles in sea eagles compared with imperial eagles. In 2004, we identified eagle species using sequences from the mitochondrial cytochrome *c* oxidase I gene (Rudnick *et al.*, 2007). We also compared genetic profiles of feather samples collected at occupied eagle territories in 1999–2002, to distinguish territory-holders from non-territorial, communal roosting non-breeders (Rudnick *et al.*, 2005). To identify origins of non-breeders we compared their genetic profiles to those of locally hatched chicks (Rudnick *et al.*, 2008).

Mark-recapture analysis

A mark-recapture study was carried out to augment and evaluate the results of the non-invasive feather collection from 2004; observational data were not used in the mark-recapture analysis. The techniques and results presented here are reported in Rudnick *et al.* (2008). In brief, once feathers were assigned to individuals, multiple feathers collected from the same individual during a single sampling occasion were collapsed into a single capture. Program MARK was used to estimate abundance of eagles under a series of closed-capture models (White & Burnham, 1999). All models were run under the Closed Population Estimation option and abundance was estimated directly. Closed-capture models were ranked by AIC_c values (Akaike's Information Criterion corrected for small sample size) and the model with the smallest AIC_c used to estimate abundance.

Demographic modelling

To understand the demographic and conservation implications of biased estimates of the size of non-breeding components of the population, we compared our genetic and observational count data to those generated by a stochastic, structured and closed demographic model of Naurzum's imperial eagles. This model is a stage structured population model of the type regularly used to understand demography and develop conservation management schemes (Katzner, Milner-Gulland & Bragin, 2007). The model reflects age-related behaviour patterns (e.g. breeding, movement) observed at Naurzum and shown by other *Aquila* eagles (Watson, 1997; Ferrer, 2001), and it reflects spatial structure in the environment by performing most calculations at the level of the four regions within the reserve. We assumed that adult birds, if present, occupy territories ahead of subadults and that birds younger than 4 years old never breed. We modelled the effects of stochastic fluctuations on eagle demography by drawing, from binomial or normal random distributions, variables describing survival, territory occupancy and reoccupancy, probability of breeding, probability of breeding success and annual chick production at successful nests. Density dependence is included in the model in an absolute way because population growth is limited by a cap on the maximum number of territories at the reserve and

because the potential for subadult breeding depends on the number of adults in the population.

Model parameters were estimated either from data collected during the long-term field monitoring described above or from a combination of published data on similar species and preliminary field data. A complete description of the model, including more detail on field techniques and parameter estimation, is provided elsewhere (Katzner, Milner-Gulland & Bragin, 2007).

We randomly picked 1000 sets of initial parameter values from a range of observed or estimated parameter values. For each set of initial values we ran our demographic model to completion 10 times, thus producing a total of 10 000 model runs. To evaluate the impact of mischaracterization of the size of the non-breeding population on modelled population estimates, we compared (1) the modelled and non-invasively estimated population sizes when the modelled number of breeders was the same as the observed number of breeders; (2) the modelled and observed number of breeders when the modelled population size was within 10% of the non-invasively estimated population size.

Results

Traditional observational counts of non-breeders averaged 20.5 ± 9.4 ($\pm$ SD; $n = 10$ from 1978 to 2004, range = 7–37). In 2003, we observed 17 non-breeding imperial eagles. In 2004, we observed 21 non-breeding imperial eagles and one non-breeding white-tailed sea eagle (Fig. 2). In 2004, we recorded occupancy by adults at 34 imperial eagle territories and we observed 38 fledged imperial eagle chicks.

We successfully extracted DNA and amplified microsatellites from 109 feathers collected in 2003 and from 1146 feathers collected in 2004 (Rudnick *et al.*, 2005, 2008). Analysis of the 109 non-invasively collected feathers from 2003 conclusively identified 47 non-breeding imperial eagles, 2.8 times more than were observed (Fig. 2). Genetic analysis of the 1146 feather

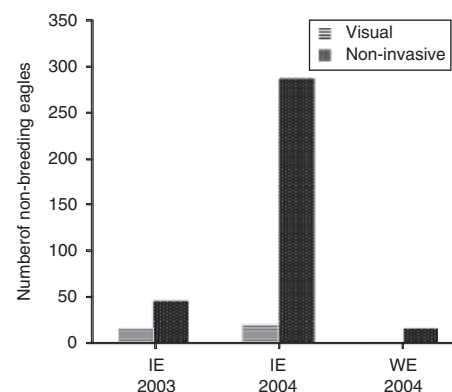


Figure 2 Visual and non-invasive counts of non-breeding imperial eagles *Aquila heliaca* (IE; 2003, 2004) and white-tailed sea eagles *Haliaeetus albicilla* (WE; 2004 only) at a communal roost at the Naurzum National Nature Reserve, Kazakhstan. Monitoring in 2004 was longer duration and more thorough than in 2003.

samples from 2004 conclusively identified 287 non-breeding imperial eagles, 13.7 times more than observed, and 16 white-tailed sea eagles, 16 times more than observed (Fig. 2). Because individual identification of sea eagles was based on a small number of microsatellite markers (i.e. the five of seven markers that amplified well for this species), this estimate is an absolute minimum count of sea eagles present. In the case of imperial eagles, > 65% of birds were identified on the basis of two shed and genotyped feathers, giving us confidence that the low error rates we report in previous studies using these techniques were also found in this study.

Genetic profiles of roost feathers did not match those of any known territorial eagles, implying that all individuals identified were from the non-breeding component of the population. Philopatry was also low; only three (6.4%) of the non-breeders identified in 2003 and 11 (3.8%) of the non-breeders identified in 2004 (Rudnick *et al.*, 2008) genetically matched profiles of chicks hatched in previous years at the reserve.

In 2004, we used repeated feather collections to conduct a mark–recapture analysis using MARK software to estimate the number of non-breeding imperial eagles using these roosts (Rudnick *et al.*, 2008). The number of eagles genetically identified during the three collections used in this analysis was 271. The most likely mark–recapture model estimated that 308 ± 8 eagles were present, 14.7 times more than were observed during any one visual survey (AIC_c value = -1440.4760 ; $n = 4$ parameters; Rudnick *et al.*, 2008), but only 21 (7%) more than were ‘counted’ genetically in all four sampling periods. Thus the mark–recapture analysis suggests that genetic sampling ‘captured’ a large component of the population.

Our demographic model was based upon best estimates of population structure from our visual observations and the literature (Katzner, Milner-Gulland & Bragin, 2007) but its output was only rarely in concordance with the non-invasive analysis of cryptic population size (Fig. 3). Of the 10 000 model runs, in 243 cases (2%) there were 34 female breeders in the model (the observed number of breeding females in 2004). However, only in 0.76% of the 10 000 cases was the modelled total population size within 10% of the total population size as estimated from mark–recapture analyses of collected feathers. The 414 imperial eagles ($n = 308$ non-breeders + 68 territory holders + 38 chicks) our genetic and observational analyses determined were using the reserve in 2004 was 326% larger than the 127 birds visually observed ($n = 21$ non-breeders + 68 territory holders + 38 chicks) and 265% larger than the population size predicted by demographic models with the same number of breeders ($n = 156 \pm 7.2$; $\pm$ SE). When the simulated population size was within 10% of the true population size (range = 339–451), the model was reasonably accurate at estimating the number of observed breeding females [$n = 33 \pm 0.47$ ($\pm$ SE), range 24–39] and the number of chicks produced [$n = 34 \pm 1.18$ ($\pm$ SE), range 14–56].

Discussion

Conservation assessments depend on biologically meaningful and statistically reasonable estimates of numbers of

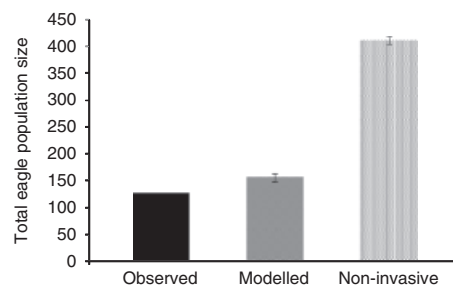


Figure 3 Estimated numbers ($\pm$ SE) of imperial eagles *Aquila heliaca* at the Naurzum National Nature Reserve, Kazakhstan in 2004. ‘Observed’ is the sum of the observed number of territorial eagles, chicks and non-breeders. ‘Modelled’ is the average number estimated in simulations based upon the observed number of breeders (which is a robust estimate because occupied territories are not missed by observers; error bars show the SE of the mean of the 243 cases considered). ‘Non-invasive’ is the total population of birds in the reserve, where the number of non-breeders is estimated by non-invasive genetic techniques and the number of breeders by observation (error bars are SEs from the MARK population estimate).

individuals. The vast majority of population estimations of threatened species are based on visual counts. In the case of globally vulnerable imperial eagles and white-tailed sea eagles, non-invasive genetic monitoring identified 13- to 16-fold more eagles than were counted with visual monitoring. Because of our low error rates, our highly conservative mechanism for handling errors and the large proportion of birds identified from multiple genetic observations, we have strong confidence in these sampling data. Thus the methods used strongly influenced and improved the estimate of population size produced. Here, traditional visual monitoring and the population estimates and modelling based on that monitoring failed to account for up to 70% of the eagles actually present. This resulted in population estimates 250–350% larger than predicted by our models. These results may not be too surprising given the known demography and itinerant pre-adult behaviour of these long-lived species. However, the Rudnick *et al.* (2008) study was the first to estimate floater numbers in populations of such a large raptor and, as we show here, there are substantial consequences of disparity between the visual and genetic methods for our understanding of eagle demography.

Although visual observations sample the number of individuals present at a particular time, our non-invasive sampling counted all birds that used a site over the course of a single breeding season. These estimates also allow us to distinguish between individuals that we counted repeatedly using the communal roost and those birds that were only counted once via a single shed feather. The low apparent philopatry and other trends reported by Rudnick *et al.* (2008) together with our new data and modelling results imply that non-breeding individuals play an important and previously unaccounted for role in the demography of Naurzum’s eagles. Because the majority of these non-breeding eagles are pre-adults that we suspect will eventually hold territories elsewhere (there are

only ~40 territories and ~300 floaters looking for nests), these results highlight this site's importance for imperial eagle conservation as a refugium for non-breeders and a source for future breeders (Ryabtsev & Katzner, 2007).

Historical monitoring of eagles at the Reserve has focused on observations of breeding that form the basis for conservation monitoring and management. Recent demographic modelling pointed to adult survivorship as a key vital rate requiring additional monitoring effort (Katzner, Milner-Gulland & Bragin, 2007). These results suggest that even those latest models are built on an already outdated understanding of eagle population dynamics that misses a crucial life stage in the population and dramatically underestimates the number of birds that use the reserve. More effective conservation monitoring would include using non-invasive genetic population estimation to provide new insight into the demography of the largest segment of the population – the cryptic non-breeders (Rudnick, DeWoody & Katzner, 2009). Effective management would then take these transient but numerous individuals into consideration.

In spite of the potentially large size of non-breeder populations, most short-term studies produce estimates for size of structured populations that do not account for cryptic non-breeders (Crouse, Crowder & Caswell, 1987; Doak, 1995; Hunt, 1998). This research shows that models and conservation management programmes built around traditional monitoring approaches can misinterpret demographic structure and potentially form a weak framework for conservation efforts. Accurate and rapid estimation of the number of cryptic non-territory holders in structured populations of long-lived species should be a priority for future research and conservation. The non-invasive techniques that we describe here and previously (Rudnick *et al.*, 2005, 2008) work well for communally roosting species, although a future priority for research should be development of similar and equally revealing techniques for less aggregated individuals.

Acknowledgements

Funding was from the Wildlife Conservation Society, the National Birds of Prey Trust, the US National Science Foundation (INT-0301905) and the National Geographic Society. E.J.M.G. acknowledges the support of the Leverhulme Trust and a Royal Society Wolfson Research Merit award. The National Aviary, Purdue University and the Naurzum National Nature Reserve provided logistical and institutional support.

References

- Crouse, D.T., Crowder, L.B. & Caswell, H. (1987). A stage-based population model for loggerhead sea turtles and implications for conservation. *Ecology* **68**, 1412–1423.
- Doak, D.F. (1995). Source-sink models and the problem of habitat degradation: general models and applications to the Yellowstone grizzly. *Conserv. Biol.* **9**, 1370–1379.
- Ferrer, M. (2001). *The spanish imperial eagle*. Barcelona: Lynx Edicions.
- Hunt, W.G. (1998). Raptor floaters at Moffat's equilibrium. *Oikos* **82**, 191–197.
- Katzner, T., Milner-Gulland, E.J. & Bragin, E.A. (2007). Using modelling to improve monitoring of structured populations: are we collecting the right data? *Conserv. Biol.* **21**, 241–252.
- Kenward, R.E., Marström, V. & Karlbom, M. (1999). Demographic estimates from radio-tagging: models of age-specific survival and breeding in the goshawk. *J. Anim. Ecol.* **68**, 1020–1033.
- Kenward, R.E., Walls, S.S., Hodder, K.H., Pakkala, M., Freeman, S.N. & Simpson, V.R. (2000). The prevalence of non-breeders in raptor populations: evidence from rings, radio-tags and transect surveys. *Oikos* **91**, 271–279.
- Martin, J., Kitchens, W.M. & Hines, J.E. (2007). Importance of well-designed monitoring programs for the conservation of endangered species: case study of the snail kite. *Conserv. Biol.* **21**, 472–481.
- Newton, I. & Rothery, P. (2001). Estimation and limitation of numbers of floaters in a Eurasian Sparrowhawk population. *Ibis* **143**, 442–229.
- Paetkau, D. (2003). An empirical exploration of data quality in DNA-based population inventories. *Mol. Ecol.* **12**, 1375–1387.
- Rudnick, J.A., Katzner, T.E. & DeWoody, J.A. (2009). Genetic analyses of noninvasively collected feathers can provide new insights into avian demography and behavior. In *Handbook of nature conservation*: 181–197. J. B. Aronoff (Ed.). Hauppauge: Nova Science Publishers.
- Rudnick, J.A., Katzner, T.E., Bragin, E.A. & DeWoody, J.A. (2007). Species identification of birds through genetic analysis of naturally shed feathers. *Mol. Ecol. Notes* **7**, 757–762.
- Rudnick, J.A., Katzner, T.E., Bragin, E.A. & DeWoody, J.A. (2008). A non-invasive genetic evaluation of population size, natal philopatry, and roosting behavior of non-breeding eastern imperial eagles (*Aquila heliaca*) in central Asia. *Conserv. Genet.* **9**, 667–676.
- Rudnick, J.A., Katzner, T.E., Bragin, E.A., Rhodes, O.E. Jr. & DeWoody, J.A. (2005). Using naturally shed feathers for individual identification, genetic parentage analyses, and population monitoring in an endangered Eastern imperial eagle (*Aquila heliaca*) population from Kazakhstan. *Mol. Ecol.* **14**, 2959–2967.
- Ryabtsev, V. & Katzner, T. (2007). Severe declines of Eastern Imperial Eagle *Aquila heliaca* populations in the Baikal region, Russia: a modern and historical perspective. *Bird Conserv. Int.* **17**, 197–209.
- Watson, J. (1997). *The golden eagle*. 1st edn. London: T. & A. D. Poyser.
- White, G.C. & Burnham, K.P. (1999). Program MARK: survival estimation from populations of marked individuals. *Bird Study* **46** (Suppl.): 120–138.
- Yoccoz, N.G., Nichols, J.D. & Boulinier, T. (2001). Monitoring of biological diversity in space and time. *Trends Ecol. Evol.* **16**, 446–453.